AN OPEN MORPHISM IS GENERIZING, AND A UNIVERSALLY OPEN MORPHISM IS UNIVERSALLY GENERIZING

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file pinned-original-source.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. Following the multi-claim convention for this source, source_url is the source-level URL https://stacks.math.columbia.edu/ and the source label lemma-open-generizing is carried in claim_key and here rather than in a per-claim tag URL.

Editorial note. The preamble is the chapter's shared Stacks Project preamble, retained as sources/stacks-morphisms/retained-preamble.tex: a line-for-line excerpt of the shared Stacks preamble staged with this snapshot. No macro was invented or redefined, and this card adds no preamble text of its own.

Editorial note. Statement span 4518-4529 and proof span 4531-4542 come from the halfA pool card operations/pool-cards/stacks-morphisms-halfA.json and were re-read in the pinned chapter file before this card was written.

Editorial note. The statement contains a reference environment recording that the result follows from the cited EGA corollary; the project's own shared preamble defines that environment as a comment environment, so it is not typeset and no bibliography is attached, exactly as in the project's own output.

Editorial note. The proof uses the author's notation for specialization arrows; it is preserved.

Editorial note. 2 cross-references occur inside the counted spans; they are listed in external_references as navigation pointers only.

Allowed standard prerequisites (editorial).

- The affine criterion for an open ring map in terms of going down for specializations, together with the translation between going down and the lifting of generalizations for the corresponding morphism of spectra, cited from the algebra chapter. Cited at proof lines 4537-4542. Gives the affine case of the statement.
- The reduction of a statement about a single specialization to an affine chart containing the two relevant points, using that the specialization of the image point stays inside an affine open containing the image. Used at proof lines 4531-4537. Lets the proof assume both schemes affine without losing the specializations involved.

Source reference. Original lemma 00HW: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 658–658. Same-chapter reference used in the retained author text at source line(s) 4487, 4541. It is supplied only as a navigation pointer so that the retained text compiles; the counted item delivers the author's

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own statement and the author's complete proof, and no part of the referenced item is copied in or substituted for the counted argument.

Source reference. Original lemma 0407: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 5246–5246. Same-chapter reference used in the retained author text at source line(s) 4539. It is supplied only as a navigation pointer so that the retained text compiles; the counted item delivers the author's own statement and the author's complete proof, and no part of the referenced item is copied in or substituted for the counted argument.

Lemma 0.1. *Let $\varphi : X \to Y$ be a morphism of schemes. If φ is open, then φ is generizing (i.e., generalizations lift along φ). If φ is universally open, then φ is universally generizing.*

Proof. Assume φ is open. Let $y' \rightsquigarrow y$ be a specialization of points of Y. Let $x \in X$ with $\varphi(x) = y$. Choose affine opens $U \subset X$ and $V \subset Y$ such that $\varphi(U) \subset V$ and $x \in U$. Then also $y' \in V$. Hence we may replace X by U and Y by V and assume X, Y affine. The affine case is Algebra, Lemma 0407 (combined with Algebra, Lemma 00HW). □